GG-L-261D
June 19, 1984
SUPERSEDING
GG-L-261C
March 3, 1977

FEDERAL SPECIFICATION

LEVELS, SURVEYING: LOCATORS, HAND-TYPE

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of hand levels for determining a horizontal line of sight.

1.2 Classification.

1.2.1 Types. Levels furnished under this specification shall be the following types, as specified (see 6.2):

Type I - Without draw and magnification of the field of view.
Type II - With draw, 2X magnification of the field of view and 4X magnification of the level vial.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

FSC 6675

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Federal Specifications:

DD-G-451 - Glass, Float or Plate, Sheet, Figured (Flat, for Glazing, Mirrors and Other Uses).
KK-L-154 - Leather, Cattlehide, (Bag and Case).
CCC-A-680 - Artificial Leather, (Cloth Coated), Vinyl Resin, Expanded Layer, (Upholstery).

Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-S-3534 - Surveying Instruments and Accessories, Packaging of.
MIL-G-20241 - Gasket Material, Wood Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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Federal Regulations:

41 CFR 1-1.25 - Recovered Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

ASTM D 822 - Operating Light- and Water-Exposure Apparatus (Carbon/Arc Type) for Testing Paint, Varnish, Lacquer and Related Products.
ASTM D 3951 - Standard Practice for Commercial Packaging.
ASTM G 23 - Light- and Water-Exposure Apparatus (Carbon-Arc Type).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description.

3.1.1 Type I level. The type I level shall consist of a tube with an eyepiece at one end and an objective lens at the other end. A spirit level shall be mounted near the objective lens on the inside, or attached on the outside, of the tube. A reflector or prism assembly, which is adjustable for the line of sight, shall be mounted inside the tube in a manner to permit viewing the image of the bubble in the field of view. Total length shall be not more than 6-1/2 inches.

3.1.2 Type II level. The type II level shall consist of a draw tube with an attached eyepiece and a tube with an objective lens. A spirit level shall be mounted near the objective lens on the inside, or attached on the outside, of the tube. A reflector or prism assembly, which is adjustable for the line of sight, shall be mounted inside the objective tube in a manner to permit sighting the bubble image in the field of view. Total length shall be not more than 6-1/2 inches with the draw tube fully retracted.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical.

3.3.2 Material deterioration and control. The level shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.2.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.2.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.3 Glass. The front panel glass shall conform to DD-G-451, type I, class 1, quality q1.

3.3.4 Leather. Leather shall conform to KK-L-154, type I or II, class 1, style optional.

3.3.5 Artificial leather. Artificial leather shall conform to CCC-A-680, class optional, treatment (b).

3.4 Tube.

3.4.1 Type I. The type I tube (round or square) shall house the eyepiece, objective lens assembly, a spirit level, and a reflector or prism assembly. The ends of the tube shall have internal threads to accommodate the eyepiece lens assembly.

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3.4.2 Type II. The type II tube assembly (round or square) shall house the draw tube with eyepiece, objective lens assembly, a spirit level, and a reflector or prism assembly. The objective tube shall have internal threads to accommodate the objective lens assembly in one end and a draw tube with eyepiece at the other end. The draw tube shall have internal threads to accommodate the eyepiece.

3.5 Eyepiece. The eyepiece shall be machined from a metal block of the same material as the tubes. One end shall have screw threads which will mate with the threaded internal end of the tubes. The aperture shall be the correct size to bring the image of the bubble and mark on the vial into sharp focus, and to achieve the accuracy required by 3.9.

3.6 Objective glass assembly. The objective lens assembly shall consist of a metal frame and objective glass. The metal frame shall be machined from a metal block compatible with the objective tube material. It shall be threaded on one end to mate with the internal threads of the objective end of the tube. The objective glass shall be securely held in the frame.

3.7 Level vial and mounting. The level vial shall be glass, shall be a suitable diameter and length, and shall have uniform curvature. The sensitivity of the bubble shall be 2 $\pm$ 0.5 millimeters (mm) of bubble movement for 30 arc minutes change in inclination. The vial shall be filled with a liquid that shall withstand the low temperature storage requirement without freezing (see 3.12). A black line shall be etched and filled or scribed and filled on the reflecting surface of the mirror or on the glass across the center of the vial. The line shall be narrow enough to achieve the accuracy required in 3.9. The vial shall be mounted so that the image of the bubble will be visible in the reflector.

3.8 Reflector. The reflector shall be a glass or plastic prism or a metal mirror capable of withstanding field conditions over a period of time without affecting the serviceability of the instrument in any way. The reflector shall be securely mounted, and the mount shall be sucurely fastened to the side of the tube. Adjusting screws or other adjusting devices shall be covered or concealed to preclude inadvertent loss of adjustment through tampering. The image of the level bubble shall be visible in the reflector.

3.9 Accuracy. When the image of the center of the bubble and the index mark of the vial are coincident, the line of sight shall not deviate from a level line more than 0.25 percent gradient.

3.10 Identification marking. The name or trademark of the manufacturer shall be stamped or otherwise permanently fixed on the instrument.

3.11 Treatment and painting. Unless otherwise specified (see 6.2), the exterior of the level shall be treated and painted in accordance with the contractor's standard practice. The inside of the tube shall be lusterless black.

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3.12 Temperature conditions. All movements of the level shall remain free and shall function satisfactorily without readjustment or change of lubricants when the level is in thermal equilibrium at any temperature from -25° F to +110° F. The level shall not be damaged by storage at an ambient temperature of +145° F for a period of not less than 4 hours. Additionally, the level shall not be damaged by storage at an ambient temperature of -28° F for a period of not less than 3 days.

3.13 Weather conditions. The level shall withstand a weathering test of 1 year or its equivalent in accordance with ASTM D 822 and ASTM G 23.

3.14 Vibration. When tested as specified in 4.5.2.5, the level shall show no signs of physical damage or loosening of any parts, including fasteners.

3.15 Carrying case. A sewed leather or artificial leather carrying case with belt loop compatible with a military belt, designed to fit and secure the hand level specified herein, shall be provided.

3.16 Workmanship. All parts, components, and assemblies of the level and case shall be clean and free of sand, dirt, fins, pits, flux, and other harmful extraneous material. All edges shall be rounded or chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (4.4).
c. Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sample shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. Upon completion of the examination specified in 4.5.1, the first article samples shall be subjected to the tests specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 0.65 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.3 and 4.5.2.4. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The levels shall be examined as specified herein for the following defects:

Major

101. Components not as specified.
102. Materials not as specified.
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, meterial finishes or treatments.
106. Objective lens loose.
107. Eyepiece lens loose.
108. Level vial not in proper place.
109. Aperture of eyepiece does not bring reflected image to bubble and mark on vial into sharp focus.
110. Carrying case not as specified.
111. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification marking missing.

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4.5.2 Tests.

4.5.2.1 Temperature.

4.5.2.1.1 Low temperature. The level shall be subjected to a temperature of -25° F long enough to permit the largest internal mass of the instrument to attain temperature stability. All movements shall then be tested. The temperature shall then be lowered until the storage temperature of -28° F has been reached, and the temperature shall be maintained for not less than 3 days. All movements shall be tested. Any evidence of damage, or nonconformance to 3.12 shall constitute failure of this test.

4.5.2.1.2 High temperature. The level shall be subjected to a temperature of +110° F long enough to permit the largest internal mass in the instrument to attain temperature stability. All movements shall then be tested. The temperature shall then be lowered until the storage temperature of -28° F has been reached, and the temperature shall be maintained for not less than 3 days. All movements shall be tested. Any evidence of damage, or nonconformance to 3.12 shall constitute failure of this test.

4.5.2.2 Weathering. The level shall be subjected to an accelerated weathering test of 1 year or its equivalent in accordance with ASTM D 822 and ASTM G 23. Evidence of rust, chalking, checking, cracking, blistering, flaking, or scaling of paint shall constitute failure of the test.

4.5.2.3 Vial sensitivity. The vial shall be mounted on a smooth horizontal beam. One end of the beam shall be supported on a pivot and the other by a micrometer screw. The beam shall be calibrated so that each revolution of the micrometer screw will result in a known angular displacement of the beam, both above and below the horizontal plane, to within 1 minute or arc. The vial bubble movement shall be measured for known angular displacements of the beam both above and below the horizontal. A sensitivity of the bubble other than 2 $\pm$ 0.5 mm of bubble movement for a 30 arc-minute change in inclination shall constitute failure of this test.

4.5.2.4 Accuracy. With the level held against a mark on a vertical post or rod, sight at a second vertical rod not less than 25 feet away. Mark point sighted. Transfer level to this mark and back sight to the first rod, again marking point sighted. Measure distance between initial and final marks and calculate error as percent gradient.

$$\text{Percent gradient} - \frac{(\text{Dist between marks})\ (100)}{(\text{Dist between rods})\ (2)}$$

A percent gradient more than 0.25 shall constitute failure of this test.

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4.5.2.5 Vibration. The level shall be vibrated at a frequency of 30 cycles per second and an amplitude of 1/16 inch in any convenient axis for a period of 15 minutes. At the end of 15 minutes, the level shall be examined as specified herein. The level shall then be tested for accuracy in accordance with 4.5.2.4. Any evidence of breakage or permanent deformation, or nonconformance to the failure criteria of 4.5.2.4 shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking for level A or B shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534. Commercial preservation, packaging, packing, and marking shall be examined to determine compliance with ASTM D 3951.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Levels shall be preserved, packaged, and packed in accordance with MIL-S-3534 for level A or B and in accordance with ASTM D 3951 for commercial. The degree of preservation and packaging shall be level A, B, or commercial and the degree of packing shall be level A, B, or commercial, as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. Packages and shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military activities. Marking for level A or B shall be in accordance with MIL-STD-129. Marking for commercial shall be in accordance with ASTM D 3951. In addition, commercial packs shall have the weight and cube data marking on the shipping container.

6. NOTES

6.1 Intended use. The level is used for determining a horizontal line of sight.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type of level required (see 1.2.1).
c. When a first article is required for inspection and approval and the number of units required (see 3.2).
d. Treatment and painting required, if other than as specified (see 3.11).
e. Degree of preservation, packaging, and packing required (see 5.1).

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6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first articles.

MILITARY INTEREST:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review activities

Navy - MC
Air Force - 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT - NHT
HHS - FHW
GSA - FSS
USDA - SCS

PREPARING ACTIVITY:

Army - ME

Project 6675-0736

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(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-L-261D	Levels, Surveying; Locators, Hand-Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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